

Response Geometry in Per-Step Integrated Gradients for Diffusion-Policy Vision-Language-Action Models

Anonymous authors

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Revision status (October 1, 2026). This working draft¹ reports analytic results and a canonical retrospective reanalysis. Actual-model numerical validation and prospective paired ranking and weight-arrangement controls remain in progress. It is not submission-ready.

Abstract

We study how scalar response choices affect the evaluation of Integrated Gradients (IG) for diffusion-policy vision-language-action models. Our application to the Robotics Diffusion Transformer attributes a fixed-noise predicted action chunk to vision and language features and proprioceptive state at each policy call. The primary scalar is an auxiliary quadratic action-discrepancy score, rather than an identified policy likelihood. A retrospective analysis of saved perturbation curves separates the ranking used to select features from the response used to score their removal. For the same recorded quadratic-target rankings, changing only the response to negative ℓ_2 changes median deletion AUC from 0.4479 to 0.2908 without changing a selected feature. Conversely, quadratic rescoring changes the recorded ℓ_2 rankings’ deletion AUC from 0.2830 to 0.4404. An equal-feature example also shows that random-order normalized AUC need not be 0.5. These results identify response geometry as a confound in cross-target attribution comparisons. We derive the exact relation between the responses and their path gradients, and construct a nonlinear policy whose IG feature ordering reverses even though the two nonzero gradients are pointwise collinear. Historical RDT experiments additionally report nonzero action displacement, persistent correlations after backbone randomization, and finite completeness residuals. They do not isolate denoiser contraction, establish the dominance of input anchoring, or certify ranking efficacy without matched controls. We distinguish reproducible saved-record analyses from historical summaries whose original inputs are unavailable, and identify numerical and provenance requirements for a controlled comparison. The public artifact provides code and saved metrics for the analyses described here.

1 Introduction

Robotic manipulation policies that fuse vision, language, and proprioceptive state have advanced rapidly. Vision-language-action (VLA) models such as RT-2 (Zitkovich et al., 2023), Octo (Octo Model Team et al., 2024), and the Robotics Diffusion Transformer (RDT) (Liu et al., 2025) execute multi-step tasks from raw sensory input conditioned on natural-language instructions. Diffusion-based policies (Chi et al., 2023) have emerged as a particularly effective family within this class. They generate action chunks through iterative denoising (Ho et al., 2020) and can model multimodal action distributions; benchmark performance depends on the task and training setup.

Despite these advances, diffusion-policy VLAs remain opaque. When a robot fails to grasp an object or misreads an instruction, diagnosing the contributions of vision, language and state is difficult from outcomes alone. This limits deployment in safety-critical settings, hampers systematic debugging, and makes it harder for operators to decide when to intervene.

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Post-hoc attribution offers a principled path toward interpretability without modifying the policy. Integrated Gradients (IG) (Sundararajan et al., 2017) attributes a model’s scalar output to its inputs by integrating gradients along a straight-line path from a baseline to the actual input. It satisfies a completeness axiom that makes the attribution quantitatively checkable. IG has been applied to image classifiers, question-answering systems and visual question answering models (Sundararajan et al., 2017; Mudrakarta et al., 2018), motivating study of the corresponding construction in diffusion-policy episodic control. Extending it to this setting raises three engineering obstacles. The denoising chain samples noise, so the attributed scalar must specify how noise is conditioned on or averaged over. The multimodal input space needs distinct baselines and interpolation strategies per modality. Episodic control demands attribution at thousands of decision points across hundreds of episodes.

This paper addresses those engineering obstacles and studies the interpretation of attribution evaluation. A perturbation curve depends on both the feature ranking and the scalar response to each perturbed input. Changing the response from a quadratic discrepancy to a negative ℓ_2 discrepancy can change normalized AUC even when the ranking and perturbed model outputs are fixed. We demonstrate this directly on saved RDT curves and give an analytic counterexample to a universal random-ranking AUC of 0.5. The question of whether a different attribution target produces a better ranking is separate and requires paired contexts, a common response, and informative controls. Likewise, a small scalar discrepancy does not imply that the predicted action is unchanged, and a low input-shuffle correlation does not establish learned-model dependence.

Our contributions are as follows.

1. **Application.** A per-policy-call, per-modality IG construction for a fixed-noise diffusion sampler, with an explicit auxiliary action-discrepancy target and modality-specific feature baselines.
2. **Response-geometry analysis.** Exact within-ranking rescoring of recorded perturbation curves, together with a normalized-response identity and a constructive IG rank reversal that separates evaluation geometry from attribution path weighting.
3. **Calibration counterexample.** An equal-contribution policy for which every ranking has continuous quadratic-response insertion and deletion AUC of $2/3$, establishing that 0.5 is not a universal chance reference.
4. **Empirical diagnostic record.** Historical completeness, perturbation, displacement, and randomization measurements, with explicit distinctions between reconstructible records, missing original artifacts, and unsupported causal interpretations.

A controlled comparison of attribution rankings remains necessary to establish their efficacy beyond response geometry and simple ranking controls.

2 Related Work

2.1 Attribution Methods

Sundararajan et al. (2017) introduced IG with the sensitivity and implementation-invariance axioms and showed that straight-line path integration satisfies both. Later work has addressed limitations of the vanilla method. Jha et al. (2021) computed attributions over neural stochastic differential equation models to produce smoother attribution maps, and later showed that shaping the injected noise further improves robustness (Jha et al., 2022). Walker et al. (2024) observed that gradients from saturation regions contribute noise and proposed Integrated Decision Gradients, which concentrates integration on the decision-relevant portion of the path. For vision transformers, metric-driven attributions (Walker et al., 2025b) and information-flow decompositions (Walker et al., 2025a) improve patch-level explanations. Guided IG (Kapishnikov et al., 2021) adapts the integration path to reduce pixel-level noise. Petsiuk et al. (2018) introduced the insertion and deletion curves we adopt as ranking-based faithfulness metrics. The reliability of perturbation metrics is itself an active concern. Hooker et al. (2019) evaluate attributions by retraining on ablated inputs. Rong et al.

(2022) show that deletion without retraining leaks information through the shape of the mask itself, and Tomsett et al. (2020) find that common saliency metrics are themselves unreliable. Our response-geometry analysis adds a diffusion-policy instance of that caution: changing the readout convention changes AUC while the recorded ranking stays fixed. Adebayo et al. (2018) proposed model and data randomization tests to assess whether saliency maps depend on learned parameters and training data. Their results caution against interpreting visually plausible maps as evidence of that dependence. Our backbone randomizations are limited tests because learned encoders and adaptors remain in place. For diffusion generative models in the image domain, cross-attention maps relate generated pixels to prompt tokens (Tang et al., 2023). This differs from attributing a scalar discrepancy of a predicted control chunk.

These methods target classifiers, language models, or image generators. We extend IG to diffusion-policy VLAs, which requires handling stochastic denoising, action-chunk outputs, and multimodal inputs inside an episodic control loop.

2.2 Vision-Language-Action Models

VLA models unify perception, language understanding, and action generation. RT-2 (Zitkovich et al., 2023) fine-tunes a vision-language model to emit discretized actions as token sequences. Octo (Octo Model Team et al., 2024) trains a generalist policy on a large multi-robot dataset with a transformer backbone. Diffusion Policy (Chi et al., 2023) frames action generation as conditional denoising diffusion (Ho et al., 2020). It produces action chunks by iterative refinement rather than single-step regression. RDT (Liu et al., 2025) scales this approach with a Diffusion Transformer (DiT) backbone, a SigLIP vision encoder (Zhai et al., 2023), and a T5-XXL language encoder (Raffel et al., 2020). RDT processes six camera slots (three camera views over a two-frame history) through SigLIP into 4,374 vision tokens. It combines them with up to 1,024 language tokens and a 128-dimensional unified state vector, and it generates 64-step action chunks through a five-step DPM-Solver++ (Lu et al., 2025) denoising schedule. This multimodal, multi-step architecture makes RDT a compelling but challenging target for attribution.

2.3 Interpretability in Robotics

Prior work on explaining robotic policies has focused on reinforcement learning agents in discrete-action settings. Greydanus et al. (2018) applied perturbation-based saliency to Atari agents, and counterfactual methods (Amitai et al., 2024) contrast the outcomes of executed and alternative actions for policy explanation. In continuous control, VisualBackProp (Bojarski et al., 2018) visualizes convolutional neural network (CNN) saliency for end-to-end driving policies. That work addresses convolutional driving models rather than a multimodal diffusion sampler. Recent VLA studies examine different explanation targets. Zhang et al. (2026) estimate visual influence using randomized perturbations and action-discrepancy responses, with explicit assumptions for relating mean-squared error to distributional divergence. VLA-Trace (Shi et al., 2026) combines representation comparisons, attention interventions and behavioral probes. IMPACT-VLA (Kim & Park, 2026) attributes phase-modality interventions to downstream task success through counterfactual closed-loop execution. Our estimand is the discrepancy of a fixed-noise predicted action chunk at a fixed context, not an explanation of eventual task success. The present study examines how response geometry affects the evaluation of IG rankings for this estimand; it does not claim to originate VLA interpretability.

3 Method

3.1 Integrated Gradients Preliminaries

Given a differentiable function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, an input $x \in \mathbb{R}^n$, and a baseline $x' \in \mathbb{R}^n$, the integrated gradient along dimension i is

$$\text{IG}_i(x, x') = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i} \Big|_{x' + \alpha(x - x')} d\alpha. \quad (1)$$

The *completeness axiom* guarantees $\sum_i \text{IG}_i(x, x') = F(x) - F(x')$. We approximate the integral with a uniform Riemann sum of m steps, giving

$$\text{IG}_i \approx \frac{x_i - x'_i}{m+1} \sum_{r=0}^m \frac{\partial F}{\partial x_i} \Big|_{x' + \frac{r}{m}(x-x')}. \quad (2)$$

This equal-weight endpoint average describes the historical implementation. Convergence requires regularity of the path integrand; finite-precision accumulation introduces additional error that increasing m does not isolate or necessarily reduce.

3.2 A Deterministic Target Through the Denoising Chain

At policy call t , write the sampler output as $g_\theta(o_t, z_t)$ for observation o_t and fixed diffusion noise z_t . We denote this predicted action chunk by μ_t for consistency with historical notation; it is a sample conditioned on fixed noise, not an identified conditional mean. The reference chunk $a_t = g_\theta(o_t, z_t)$ is computed once without gradients. For an intervened input, the primary attribution target is the auxiliary score

$$Q_t = -\frac{1}{2\sigma^2 D} \|a_t - \mu_t\|^2, \quad (3)$$

with $\sigma^2 = 1$ and $D = 512$ active entries: 64 predicted time steps over eight active dimensions. This is the nonconstant part of a Gaussian log kernel averaged over entries, with its log-normalizing constant omitted. It has not been derived as the diffusion policy’s conditional log likelihood. Historical code and files call this target `logpi`; we use ΔQ for its change. Attribution concerns the complete predicted chunk, of which the execution loop applies a subsampled prefix before replanning or termination.

We hold the noise realization fixed across the interpolation path to evaluate one deterministic scalar function. Attribution of an expectation over noise would be a different estimand and requires a justified gradient-expectation interchange and Monte Carlo treatment. Independent noise at each path point does not compute the fixed-noise quantity considered here. The historical implementation replaced nonfinite gradient entries by zero. That operation can corrupt individual coordinates while cancellation leaves the scalar completeness residual small or even zero; completeness is not a certificate of coordinate accuracy.

3.3 Alternative Scalar Targets

We compare the quadratic target with two alternative action-discrepancy targets,

$$y_t^{\ell_2} = -\sqrt{\|a_t - \mu_t\|_2^2 + \epsilon}, \quad y_t^{\max} = -\max_j |a_{t,j} - \mu_{t,j}|, \quad (4)$$

where $\epsilon = 10^{-12}$ is the implemented stabilizer and both targets use the same D active entries. The stabilized ℓ_2 target is smooth in the residual and has zero gradient at exact agreement; away from the stabilizer scale its action-space gradient approaches unit norm. The maximum-deviation target, abbreviated `maxdev`, uses a different norm and is not a monotone transform of ℓ_2 in general: residuals $(1, 1)$ and $(1.1, 0)$ reverse order. The quadratic and stabilized ℓ_2 responses are algebraically related for a fixed perturbation, although their gradients can induce different attribution rankings. Their units and scales differ, so a larger magnitude alone does not demonstrate greater information or operational usefulness. Section 5.2 isolates response changes for the quadratic and stabilized ℓ_2 targets. The maximum-deviation records remain an auxiliary diagnostic in the released analysis.

3.4 Response Transforms and Attribution Path Weights

Fix the reference action, active coordinates, model, noise and other modalities, and write $r(x) = \|a - \mu(x)\|_2^2$ and $L(x) = -\sqrt{r(x) + \epsilon}$. Where the gradients exist, the chain rule gives the exact identity

$$\nabla Q(x) = \frac{\sqrt{r(x) + \epsilon}}{D\sigma^2} \nabla L(x). \quad (5)$$

Thus nonzero pointwise gradients have the same direction. At exact self-reference both vanish. Along $x(\alpha) = x' + \alpha(x - x')$, however,

$$\text{IG}_i^Q = (x_i - x'_i) \int_0^1 \frac{\sqrt{r(x(\alpha)) + \epsilon}}{D\sigma^2} \partial_i L(x(\alpha)) d\alpha. \quad (6)$$

The positive weight generally varies with α . When coordinate contributions vary differently along the path, their integrated rankings can therefore change. This does not contradict IG implementation invariance, which compares implementations of the same scalar function (Sundararajan et al., 2017).

A concrete example uses the scalar action $\mu(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline $(0, 0)$, actual input $(1, 1)$, detached reference $a = \frac{9}{5}$, and $D = \sigma^2 = 1$. On the same straight path, exact quadratic-target integration yields

$$\text{IG}^Q = \left(\frac{23}{30}, \frac{64}{75} \right), \quad \lim_{\epsilon \downarrow 0} \text{IG}^L = \left(1, \frac{4}{5} \right). \quad (7)$$

Quadratic IG ranks feature 2 first, whereas stabilized L with the implemented $\epsilon = 10^{-12}$ ranks feature 1 first with a margin at least 0.1999975. Appendix A proves the finite-stabilizer bound, so the reversal does not rely on replacing the implemented response by its limit. This establishes existence in a smooth nonlinear policy, not a universal rank change or an empirical RDT effect.

3.5 Per-Step Episode Attribution

RDT outputs action chunks of 64 steps, subsampled $4\times$ to 16 environment steps. We run IG once per *policy call*, not per environment step, because the chunk is the atomic unit of decision. The attribution unit is therefore one policy query. Runtime depends on the implementation, hardware, precision and integration budget; the earlier timing summaries are not used as a performance result.

At each policy call we compute three independent attributions through modality-specific forward functions.

- **Vision.** Freeze language and state and interpolate the adapted image tokens from the baseline to the real observation.
- **Language.** Freeze vision and state and interpolate the adapted language tokens from a minimal two-token baseline encoded through T5-XXL to the real instruction embedding.
- **State.** Freeze vision and language and interpolate the 128-dimensional unified state vector (eight active dimensions for the Franka arm, namely seven joints plus the gripper) from zeros to its real value. The state-adaptor multilayer perceptron (MLP) sits inside the forward function to preserve per-joint structure.

Algorithm 1 summarizes the procedure.

Baseline selection. The vision baseline is a uniform gray image in the external-camera slot only. The remaining five camera slots carry constant background images and are frozen at their real values, so IG attributes only to the informative slot. The language baseline is a minimal two-token sequence (the tokenizer’s padding and end-of-sequence tokens, since T5 has no beginning-of-sequence token) encoded through T5-XXL rather than a zero vector. The encoder output is zero-padded to the stored embedding length, and the language forward passes attend over the task’s real token span. The attended baseline is therefore the encoded token pair followed by zero vectors at the remaining attended positions. A zero vector is numerically permitted by the GELU-based language adaptor but defines a different representation baseline; preliminary efficacy comparisons are not independently reconstructible. The state baseline is a zero vector in the normalized joint space, a numerically safe choice that keeps per-joint attributions readable. Section 5.4 quantifies sensitivity to the vision baseline choice.

Algorithm 1 Per-step IG for diffusion-policy VLAs.**Input.** Policy π_θ , environment $\mathcal{E}$, integration steps m , target $y \in \{Q, \ell_2, \text{maxdev}\}$ **Output.** Attributions $\{A_t^V, A_t^L, A_t^S\}$ at every policy call.

```

1:  $o_0 \leftarrow \mathcal{E}.\text{reset}()$ 
2: for policy call  $t = 0, 1, \dots$  do
3:   Encode  $o_t$  into vision  $v_t$ , language  $l_t$ , state  $s_t$ 
4:   Draw and store  $z_t$ ; compute  $a_t \leftarrow g_\theta(o_t, z_t)$  without gradients
5:   Define  $F$  from  $a_t$ , the fixed-noise predicted chunk  $\mu_t$ , and the chosen target
6:   for modality  $M \in \{V, L, S\}$  do
7:     Set baseline  $x'_M$ , input  $x_M$ 
8:     Freeze the other two modalities
9:      $A_t^M \leftarrow \text{IG}(F_M, x_M, x'_M, m)$ 
10:  end for
11:  Record  $A_t^V, A_t^L, A_t^S$  and completeness errors
12:  Apply action chunk and advance  $\mathcal{E}$  by up to 16 steps, stopping on termination
13: end for

```

Eq. 2

3.6 Evaluation Protocol

Completeness. For each modality, with the other inputs frozen, we report the absolute residual $|\sum_i \text{IG}_i - (F(x) - F(x'))|$ and its relative version

$$\text{Err} = \frac{|\sum_i \text{IG}_i - (F(x) - F(x'))|}{|F(x) - F(x')|}. \quad (8)$$

An exactly zero signed gap makes the relative quantity undefined. A small nonzero gap may amplify numerical error. The historical action-norm filter $\|a_t\| \geq 15$ does not bound this denominator, and is retained only as a population sensitivity. A low residual checks a scalar accounting identity; canceling coordinate errors can still change the attribution map or ranking.

Perturbation curves. We rank eligible feature groups by the sum of absolute coordinate attributions. The groups are external-camera patch tokens for vision and attended instruction tokens for language. Deletion replaces a prefix of the ranking with baseline features; insertion replaces a prefix of a baseline input with original features. For a scalar response R , the normalized ordinate is

$$\tilde{R}(x_k) = \frac{R(x_k) - R(x')}{R(x) - R(x')}. \quad (9)$$

We integrate on each record’s stored fraction grid. Neither the curve nor its AUC is necessarily bounded by $[0, 1]$. The historical grid requests $\{0, 1, 5, 10, 20, 30, 50, 75, 100\}\%$ of groups; rounding changes realized language-token fractions, so the saved curves cannot be treated as uniformly sampled continuous functions. Top- k score changes are $R(x_k) - R(x)$, with units defined by R . We distinguish the attribution target that generates a ranking from the response that evaluates its interventions. The earlier study-specific thresholds are not used as efficacy tests in the revised tables.

Randomization and baseline diagnostics. C1 reinitializes model parameters and correlates the resulting attributions with the originals. The standard variant changes the final action-output projection and recomputes its reference action; frozen-reference and backbone-cascade variants retain the original reference. The historical cascade simultaneously reinitializes all backbone parameter matrices of rank at least two; it retains one-dimensional parameters, encoders and adaptors. It is not a progressive layerwise cascade. C2 shuffles both external-camera pixels and language. The historical implementation uses a pre-T5 shuffled-language cache when available and otherwise permutes post-T5 embeddings; the saved records do not authenticate which mode each call used. These are different dependence tests, and C2 is not the label-randomization experiment of Adebayo et al. (2018). Historical perturbed maps use $m = 16$ and original maps nominally use $m = 64$, so C1 also changes the numerical approximation. A baseline sensitivity compares black, gray and blurred observation features.

4 Retrospective Data and Statistical Protocol

Scope and traceability. The revised analysis authenticates 99 preserved files containing 22,966 valid records and 1 specifically identified corrupt line. Each valid physical record has a source path, original line number and byte hash. The raw inventory contains 129 repeated-key groups, of which 58 differ in scientific fields rather than only timing metadata. The last occurrence is a declared retrospective selection convention, not evidence that a conflicting attempt is correct. A separate sensitivity removes every key with a conflicting payload. The records, occurrence decisions, population membership, result identifiers and generated manuscript cells are linked in the released lineage registry. This authenticates preserved data bytes, not missing checkpoints, observations, action tensors or noise.

Campaigns and models. The principal reconstructed RDT-170M tables use the saved Month 7 verification campaign at $m = 64$, covering PickCube, StackCube, PegInsertionSide and PickSingleYCB in ManiSkill3 (Tao et al., 2024). These are separate experiments from the unavailable original main-pass inputs. The model was pretrained RDT-170M; the saved RDT-1B PickCube campaign used a task-finetuned checkpoint. The model comparison changes training, trajectories and historical preprocessing as well as parameter count, and is not a scale or competence intervention. Table 3 therefore reports the 1B cohort separately. No missing original completeness, standard-sanity, or early PickCube/StackCube integration-budget result is copied into a primary table.

Known protocol limits. The historical action chunk is subsampled before simulator execution, while attribution concerns the full predicted chunk. The four-task scope reflects historical implementation choices. An attempted TurnFaucet run used an incorrect controller, and no corrected rerun is documented; unsuccessful outcomes under that configuration cannot justify a capability exclusion. Early follow-up jobs used a zero language baseline in place of encoded padding/end tokens, and the PickSingleYCB follow-up used a proxy PickCube instruction embedding. The pooled 1B language cohort spans the early baseline and later encoded baselines. Early sidecar filenames also omitted seed and integration budget, permitting overwrites and possible reuse across nominal runs. Seed-tagged Month 7 records mitigate that filename issue but do not recover old contexts. We do not infer exact context equality from matching scalar fingerprints.

Estimands and uncertainty. The primary population is all valid saved calls after the declared occurrence rule. Continuous call-level summaries use medians over calls; residual pass rates use proportions over the stated call population. Solver contrasts report relative changes between endpoint medians, and the separately named equal-episode analysis uses episode medians. We resample entire recorded episode groups, keyed by task, model, recorded seed and episode, with $B = 10,000$ draws and seed zero. Point estimates and intervals use exactly the same source-record population. These percentile intervals describe the released empirical episode mixture, with longer episodes contributing more calls; they do not estimate variability across independently trained checkpoints. A separately named 1B sensitivity reports the median of episode medians, weighting episodes equally. For the old randomization variants we also group potentially reused nominal task/episode keys across seeds as a conditional dependence sensitivity. Neither grouping repairs unobserved context identity. Very small cluster counts are explicitly disclosed, and two-group baseline resampling is not presented as calibrated interval coverage. All unfiltered results and the historical norm-selected sensitivity remain available; no favorable tail is removed to obtain a threshold verdict.

5 Reconstructed Retrospective Results

5.1 Completeness and Perturbation Summaries

Table 1 reports completeness residuals from the saved verification campaign. Vision and language have smaller median relative residuals than state in these cohorts, but this does not establish more accurate coordinates or a causal explanation based on dimension count. The historical integrator combines model-forward, path, summation and quadrature errors. Actual-model validation must separate those factors on identical inputs before selecting production settings.

Task	Calls	Episodes	Vision (%)	Language (%)	State (%)
PegInsertion	700	100	2.17 [1.94, 2.35]	1.83 [1.69, 1.98]	3.48 [3.00, 4.04]
PickCube	397	100	2.20 [1.90, 2.43]	1.88 [1.62, 2.30]	5.88 [4.99, 7.29]
PickYCB	397	100	1.85 [1.71, 2.06]	2.02 [1.79, 2.18]	4.26 [3.50, 5.62]
StackCube	400	100	2.39 [2.18, 2.59]	1.76 [1.58, 1.97]	10.53 [8.07, 13.13]

Table 1: Month 7 verification campaign, all valid calls, $m = 64$. Cells are median relative completeness residuals in percent with conditional 95% episode-bootstrap intervals. Calls and episodes are the identical point/interval populations. These are historical numerical diagnostics, not coordinate-accuracy certifications.

Table 2 reports native quadratic-response insertion and deletion AUCs for the same verification campaign. These are descriptive perturbation results. Without matched random and simple ranking controls, their magnitudes do not establish attribution efficacy. Normalization can produce extreme negative values: the smallest saved PickCube vision insertion AUC in the all-valid population is -18190.36. The analysis retains that observation and its endpoint denominator instead of clipping it or treating the action-norm filter as a validity certificate. Medians are consequently accompanied by the released denominator/tail diagnostics; a central summary alone does not describe the entire curve population.

Task	Modality	Calls	Episodes	Insertion AUC	Deletion AUC
PegInsertion	Vision	700	100	0.731 [0.688, 0.774]	0.280 [0.234, 0.309]
PegInsertion	Language	700	100	0.320 [0.257, 0.379]	0.341 [0.291, 0.382]
PickCube	Vision	397	100	0.822 [0.771, 0.838]	0.167 [0.137, 0.213]
PickCube	Language	397	100	0.525 [0.468, 0.580]	0.433 [0.391, 0.475]
PickYCB	Vision	397	100	0.844 [0.808, 0.863]	0.337 [0.296, 0.372]
PickYCB	Language	397	100	0.335 [0.227, 0.409]	0.471 [0.438, 0.500]
StackCube	Vision	400	100	0.821 [0.777, 0.849]	0.208 [0.161, 0.261]
StackCube	Language	400	100	0.359 [0.292, 0.432]	0.480 [0.429, 0.512]

Table 2: Month 7 verification campaign, all valid calls and the native quadratic response. Medians and conditional 95% intervals share one population. No entry is labeled a success against an uncalibrated chance threshold. The recorded token-fraction grids, endpoint conventions and provenance limits remain part of these descriptive estimates.

The saved 1B analysis illustrates a second distinction: changing the episode weighting changes the estimand. Table 3 uses the same 1,199 calls from 60 episodes under both summaries. The language summaries are more sensitive to this weighting than the vision summaries, within a cohort that also mixes language-baseline provenance. Neither estimate is a training-replication result or a controlled improvement over 170M.

Estimand	Modality	Insertion AUC	Deletion AUC
Median over calls	Vision	0.940 [0.926, 0.954]	0.288 [0.250, 0.312]
Median over calls	Language	0.541 [0.449, 0.605]	0.267 [0.187, 0.327]
Median of episode medians	Vision	0.940 [0.927, 0.959]	0.289 [0.235, 0.313]
Median of episode medians	Language	0.488 [0.368, 0.581]	0.297 [0.151, 0.342]

Table 3: Saved RDT-1B PickCube campaign, 1,199 source calls and 60 episodes under both estimands. The first weights each call equally; the second weights each episode’s median equally. Conditional 95% episode intervals describe the released mixture. Evaluation seeds share a checkpoint, and the pooled language population spans two baseline provenances.

5.2 Separating Attribution Target from Evaluation Response

The same recorded interventions can be rescored without recomputing IG or changing a selected feature. Let d be the recorded active-action discrepancy, $Q = -d^2/1024$, and $L = -\sqrt{d^2 + \epsilon}$ with $\epsilon = 10^{-12}$. Then

$$L = -\sqrt{-1024Q + \epsilon}, \quad Q = -\frac{L^2 - \epsilon}{1024}. \quad (10)$$

For an exact self-reference actual endpoint and an all-baseline squared residual $R > 0$, the normalized responses satisfy

$$\begin{aligned} N_Q(r) &= 1 - \frac{r}{R}, & N_L(r) &= \frac{\sqrt{R + \epsilon} - \sqrt{r + \epsilon}}{\sqrt{R + \epsilon} - \sqrt{\epsilon}}, \\ N_Q(r) - N_L(r) &= \frac{(\sqrt{r + \epsilon} - \sqrt{\epsilon})(\sqrt{R + \epsilon} - \sqrt{r + \epsilon})}{R}. \end{aligned} \quad (11)$$

Consequently $N_Q > N_L$ for $0 < r < R$, with equality at both endpoints and reversed ordering for $r > R$. Integration with nonnegative weights preserves the weak ordering if every intervention remains in range, even for a nonmonotone curve; strict area inequality requires positive integration weight on an interior residual. Overshooting points can change the area ordering and must not be clipped; if $R = 0$, both normalizations are undefined. The retrospective computation below transforms the recorded endpoints directly and does not assume exact historical self-reference. We transform every recorded curve ordinate, evaluate its endpoints under the transformed response, and reintegrate on the same saved grid. Each ranking cohort contains 750 calls from 30 episodes after duplicate handling. For quadratic-target vision rankings, median deletion AUC changes from 0.4479 under Q to 0.2908 under L . For stabilized- ℓ_2 rankings it changes from 0.2830 under L to 0.4404 under Q . Table 4 and Figure 1 include both modalities and both perturbation directions.

Ranking target	Response	Modality	Insertion AUC	Deletion AUC
Q	Q	Vision	0.883 [0.845, 0.909]	0.448 [0.408, 0.504]
Q	Q	Language	0.580 [0.510, 0.658]	0.372 [0.311, 0.428]
Q	L	Vision	0.756 [0.718, 0.785]	0.291 [0.253, 0.333]
Q	L	Language	0.469 [0.412, 0.516]	0.257 [0.228, 0.284]
L	Q	Vision	0.884 [0.833, 0.921]	0.440 [0.402, 0.501]
L	Q	Language	0.635 [0.582, 0.720]	0.368 [0.302, 0.407]
L	L	Vision	0.756 [0.700, 0.803]	0.283 [0.252, 0.327]
L	L	Language	0.507 [0.456, 0.566]	0.249 [0.202, 0.278]

Table 4: Within-ranking response transformation, with median AUC and conditional 95% episode-bootstrap intervals. Each ranking cohort contains 750 calls from 30 episodes. The same recorded interventions are reused within a ranking cohort. The two ranking cohorts are not authenticated paired contexts; comparisons between them do not identify ranking improvement.

This establishes response-geometry sensitivity within each saved ranking cohort. It does not establish that Q-IG and L2-IG select the same features, that their rankings are equivalent, or that either is effective. The attempted cross-cohort join has different reference-action norms at every nominal key; exact raw contexts and noise are unavailable. Apparent agreement after common rescored therefore cannot replace a prospective paired ranking comparison. Likewise, the larger native L2 score is not by itself recovered information, and stabilized L2 has no square-root kink at exact agreement.

5.3 An Analytic Random-Order Counterexample

For an additive scalar policy with equal feature contributions, an all-one input and a zero baseline, every feature ordering is identical for perturbation evaluation. At deleted fraction q , quadratic-response normalization gives $1 - q^2$; insertion gives $1 - (1 - q)^2$. Both continuous AUCs are $2/3$, despite the absence of ranking information. Trapezoidal integration on the historical requested grid gives 0.65976 for both curves. The

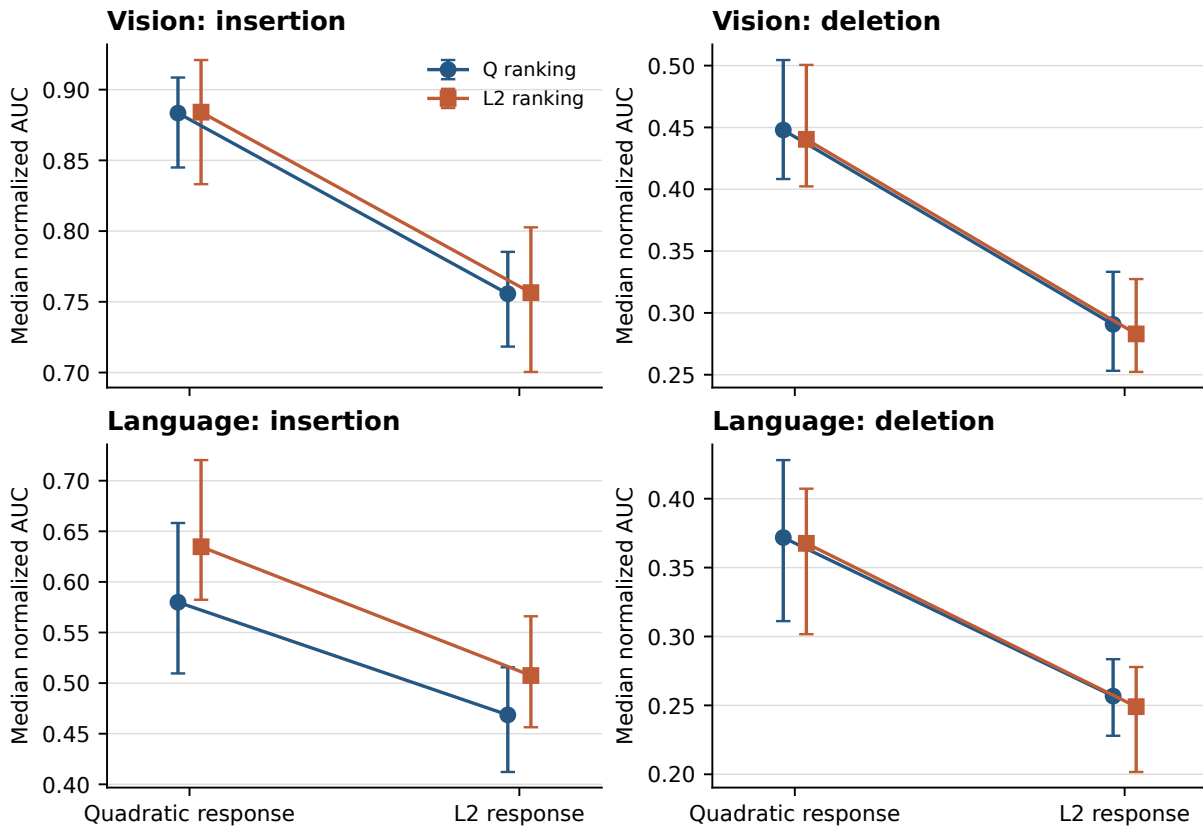


Figure 1: Changing only the evaluation response while keeping each recorded ranking and its interventions fixed. Error bars are conditional episode-bootstrap intervals for each median. Lines connect responses within a ranking cohort; the two ranking cohorts are not authenticated pairs. There is no universal chance reference line.

continuous integral and sampled-grid statistic are distinct quantities. An unstabilized negative-norm response instead produces linear normalized curves in this example. Consequently, a fixed AUC reference such as 0.5 requires justification for the policy, response and intervention protocol; it is not universal. This construction is an analytic control, not an empirical random-ranking measurement on RDT.

5.4 Randomization and Baseline Diagnostics

Table 5 reconstructs the standard verification-campaign correlations. Model randomization retains substantial rank association, while input shuffling changes the maps more strongly. That contrast does not make model dependence unnecessary or certify semantic grounding. A map based only on input magnitude can change under input shuffling while ignoring learned policy weights. The saved frozen-reference and cascade variants, including a dependence-grouping sensitivity, are reported in Appendix B.

Task	Check	Calls	Episodes	Vision ρ	Language ρ
PegInsertion	C1	100	16	0.960 [0.949, 0.969]	0.959 [0.947, 0.966]
PegInsertion	C2	100	16	0.089 [0.067, 0.111]	0.018 [0.004, 0.036]
PickCube	C1	100	26	0.956 [0.948, 0.961]	0.926 [0.917, 0.944]
PickCube	C2	100	26	0.142 [0.128, 0.157]	0.117 [0.099, 0.150]
PickYCB	C1	100	26	0.943 [0.930, 0.949]	0.917 [0.903, 0.931]
PickYCB	C2	100	26	0.164 [0.149, 0.175]	0.128 [0.081, 0.155]
StackCube	C1	100	26	0.958 [0.952, 0.965]	0.965 [0.960, 0.969]
StackCube	C2	100	26	0.140 [0.117, 0.154]	0.266 [0.246, 0.279]

Table 5: Month 7 model randomization (C1) and input shuffle (C2), all recorded calls. Conditional 95% episode intervals accompany median Spearman correlation. These diagnostics test different dependencies: low C2 correlation does not establish learned-model dependence, and C1 changes integration budget as well as weights.

The IG multiplier ($x_i - x'_i$) permits shared structure but does not force a particular ranking. The path gradients can dominate the result, and the input features themselves come from retained learned encoders/adaptors. Persistent correlations therefore motivate input-only and path-gradient-only controls; they do not establish multiplier dominance. Spearman correlation is not a fraction of model-independent information. The baseline comparisons in Appendix B show imperfect agreement but cannot establish equivalence with their small number of groups.

5.5 Action Displacement and Solver Resolution

The quadratic score is a normalized squared displacement, so its numerical magnitude has a direct dependence on the chosen convention. Removing $1/D$ multiplies exact scores and attributions by D without changing exact ranks, normalized AUC, or relative completeness. This algebra does not determine whether an action change is operationally important; that requires action units and a justified behavioral criterion.

The saved solver analysis compares median relative action displacement under top-5% vision deletion between $T = 2$ and $T = 20$. Figure 2 reports the endpoint contrast with conditional resampling of nominal episode pairs. Reference actions vary across solver settings, and exact contexts/noise cannot be authenticated. The intervals permit appreciable increases and decreases. Changing T changes discretization over the full denoising horizon, rather than adding steps along one fixed trajectory. These observations therefore identify neither a causal contraction mechanism nor its absence.

6 Numerical Validation and Prospective Study

Status of this working draft. The revised runtime’s actual-model numerical campaign is in progress. Its results and the ensuing ranking study have not yet been integrated into this manuscript. The retrospective results above do not certify a production numerical setting or close the efficacy question.

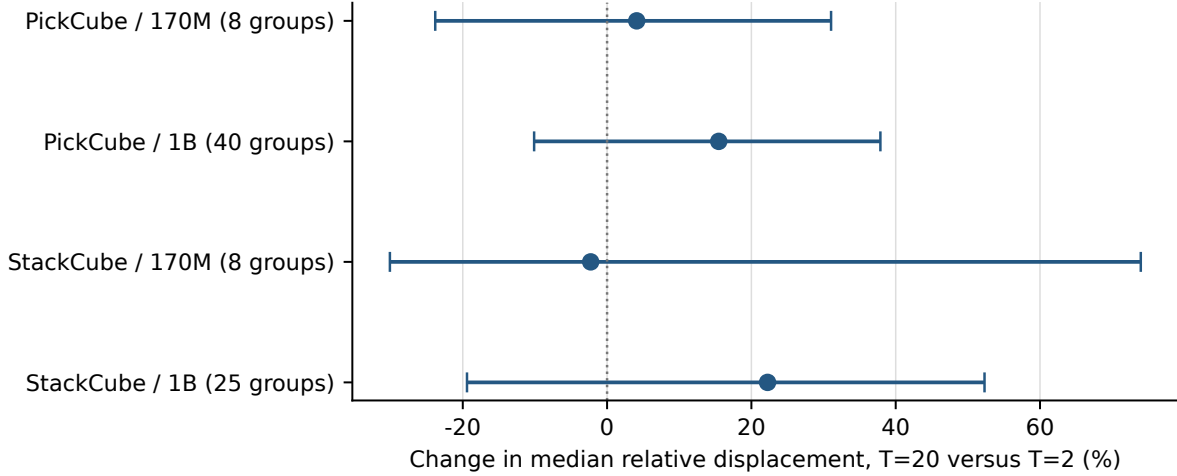


Figure 2: Retrospective change in median relative displacement under top-5% vision deletion. Intervals use nominal episode pairing across solver resolutions, conditional on the released records and grouping. The vertical reference denotes no endpoint change, not a contraction test or an equivalence margin. Raw context/noise identity remains unknown.

The numerical design uses authenticated stored observations, masks, reference actions and exact noise arrays. It separates accumulation arithmetic, master-path/product arithmetic, quadrature and downstream model precision, and compares complete coordinate arrays, eligible-group ranks, top-5% membership and perturbation responses. A downstream fp32 control lifts the same bf16-rounded parameters and cached adapted representations, explicitly converts non-parameter dtype attributes, and holds the original reference action fixed. This tests downstream arithmetic on fixed numeric inputs; it is not an end-to-end fp32 encoder study or recovery of unrounded checkpoint parameters. Repeated endpoints alone do not certify gradient/path repeatability, so the validity assessment must also inspect attribution repeats and failures. The prospective function is a separately specified deterministic fp32 probe: it re-adapts the raw state, computes its own fp32 reference, and evaluates saved contexts visited by the bf16 policy. It is an offline intervention function, rather than an exact explanation of the bf16 policy’s executed commands. Its integration procedure requires separate engineering selection and verification on fresh episodes with uniformly sampled policy calls over the complete trajectory.

After numerical settings are independently validated, a prospective study must generate Q-IG, L2-IG, random, input-difference-magnitude and path-gradient-only rankings on the same contexts and stored noise. Rankings select their own masks under common eligible groups, baselines and realized-fraction grids; every selected mask is evaluated with the same action function, reference and noise. The primary response is raw positive root-mean-square action discrepancy, integrated without endpoint normalization over realized fractions; normalized quadratic and negative-norm responses are secondary geometry diagnostics. The path-gradient-only control omits the input multiplier; the gradient at the self-reference endpoint alone is uninformative because both discrepancy gradients vanish there. Paired effects must be summarized at the episode unit, with task/checkpoint strata, declared failure denominators and precision-based sample planning. A variance-only pilot can select the episode budget without using signed comparative means for allocation; this is not a claim that the pilot is blinded. Both informative negative and positive results are relevant; neither a favorable threshold crossing nor a single engineering context is a stopping criterion for the scientific study.

A selected additional control permutes entries independently within each backbone linear weight tensor while preserving its learned value multiset. Biases, normalization parameters, positional structure, encoders and adaptors remain fixed, so this tests dependence on learned weight arrangement conditional on those retained structures. Each permuted function supplies its own action reference for attribution, and the resulting rankings are transferred to the unchanged trained fp32 response. Native initialization is a different control: the pinned RDT constructor initializes its final output head to zero, making its finite fixed-noise

output independent of conditioning and its own-reference gradients uninformative. Permutation draws will be independent across reset episodes and coherent across selected calls and modalities within an episode. The base comparison and this control retain separate joint-complete episode populations; failures in the control do not remove episodes from the base analysis. Numerical validation, parameter Monte Carlo uncertainty, simultaneous inference and precision-based allocation remain prerequisites for reporting this study’s results. An additional descriptive sensitivity will compare the full predicted chunk with its positions corresponding to commands submitted in the source bf16 rollout before replanning or termination, using the same saved fp32 intervention actions and preserving the primary population. Those counterfactual actions were not themselves submitted to the environment.

7 Discussion and Limitations

The reconstructible evidence separates three questions: what a scalar response measures, whether a numerical attribution is reliable, and whether its ranking contains useful model-dependent information. The saved response transformation answers the first question for fixed recorded interventions. It cannot answer the latter two by itself. A stronger paper requires the controlled numerical and ranking studies above rather than treating retrospective limitations as the final contribution.

Historical uncertainty also has several distinct sources. Source-record hashing can authenticate released bytes while checkpoint and context identities remain unknown. Episode resampling addresses within-episode dependence conditional on a particular empirical mixture; it does not repair shared or overwritten contexts, preprocessing confounds, or missing training replicates. The 1B results come from one checkpoint, and the solver and alternative-target analyses cover narrower task populations than the verification campaign. Normalized response curves may be unbounded and sensitive to endpoint gaps. Camera-token and language-token interventions differ in group counts and semantics, so their AUCs are not directly interchangeable measures of modality importance.

Historical overlay panels are preserved as artifacts but are not used as evidence of successful grounding or behavioral failure diagnosis. Likewise, groups previously named from completeness residuals are numerical diagnostic groups, not independently validated labels such as misgrounding or delayed credit. The unrecovered original main-pass and early budget-extension numbers are excluded from revised primary tables. Appendix C reports only reconstructible later budget cohorts, without a convergence claim.

8 Conclusion

Per-policy-call IG for a fixed-noise diffusion sampler can be defined using an explicit auxiliary action-discrepancy score. For saved RDT perturbation curves, changing only the evaluation response changes normalized AUC without changing the selected features. An analytic example rules out a universal random-ranking AUC of 0.5. The canonical retrospective tables preserve what the available records support and exclude unrecovered primary constants. Numerical validity and efficacy relative to matched controls remain requirements for the ongoing prospective revision.

Ethical Statement

This work analyzes pretrained robotic manipulation policies in simulation; no human subjects were involved. Attribution is a diagnostic tool here, not a safety guarantee for robot deployment. Misinterpreting attractive maps, low completeness residuals or uncalibrated AUCs could create unjustified confidence in a policy, which motivates the explicit limits and controls.

Reproducibility Statement

The retrospective analysis parses a hash-authenticated raw inventory, records occurrence decisions, and links each result to physical source records. The revised tables and figures are generated from the same canonical artifact, with identical point/interval populations and a manuscript lineage registry. It retains

conflicting-key exclusions, action-norm strata, endpoint/tail diagnostics, alternative episode weightings and conditional grouping sensitivities. These operations reproduce arithmetic over saved records; they do not recover missing models, sidecars or original observations. The new runtime separately records immutable context, source, checkpoint, language, environment and noise identities, and fails explicitly on nonfinite attribution rather than sanitizing coordinates. A candidate PDF or a successful script run alone does not constitute an authenticated scientific replication. The identified repository URL is omitted from this draft. An anonymous review supplement must be assembled and inspected after the scientific gates are complete.

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A Response-Geometry Derivations

The identities in Section 3.4 assume that the relevant path gradients are integrable and hold the action reference fixed. Differentiating $Q = -r/(2D\sigma^2)$ and $L = -\sqrt{r+\epsilon}$ gives Equation 5 immediately. For Equation 11, set $a_0 = \sqrt{\epsilon}$, $b = \sqrt{R+\epsilon}$ and $z = \sqrt{r+\epsilon}$. Since $R = b^2 - a_0^2$ and $r = z^2 - a_0^2$, rationalization gives

$$\frac{b^2 - z^2}{R} - \frac{b - z}{b - a_0} = \frac{(z - a_0)(b - z)}{R}. \quad (12)$$

For in-range residuals the maximum difference is $(b - a_0)^2/(4R) \leq 1/4$. These pointwise statements concern a fixed ranking’s interventions; they neither order different rankings nor establish task-success improvements.

For the rank-reversal example, let $c = 4/5$ and parameterize the path by (t, t) . The scalar residual is $d(t) = 1 + c - t^2 - ct = (1 - t)(1 + t + c) \geq 0$. The quadratic path integrands are $2td(t)$ and $cd(t)$, yielding

$$\text{IG}^Q = \left(\frac{1}{2} + \frac{c}{3}, \frac{2c}{3} + \frac{c^2}{2} \right) = \left(\frac{23}{30}, \frac{64}{75} \right). \quad (13)$$

For the stabilized negative norm the respective integrands are $2t d(t)/\sqrt{d(t)^2 + \epsilon}$ and $c d(t)/\sqrt{d(t)^2 + \epsilon}$. Using $d(t) \geq c(1 - t)$ and monotonicity of $1 - d/\sqrt{d^2 + \epsilon}$ gives

$$\int_0^1 \left(1 - \frac{d(t)}{\sqrt{d(t)^2 + \epsilon}} \right) dt \leq 1 - \frac{\sqrt{c^2 + \epsilon} - \sqrt{\epsilon}}{c} \leq \frac{\sqrt{\epsilon}}{c}. \quad (14)$$

Therefore $0 \leq 1 - \text{IG}_1^L \leq 2\sqrt{\epsilon}/c = 2.5 \times 10^{-6}$ and $0 \leq c - \text{IG}_2^L \leq \sqrt{\epsilon} = 10^{-6}$ at the implemented stabilizer. All coordinates are positive and the stabilized L margin is at least 0.1999975, while Q has the opposite ordering with exact margin 13/150. The artifact includes independent differentiation, integration and normalization checks for these identities, including overshoots, zero gaps and a policy example without rank reversal.

B Additional Retrospective Diagnostics

The frozen-reference and backbone-cascade analyses retain learned encoders/adaptors and historical context uncertainty. Table 6 reports both recorded-episode grouping and a conditional shared-context grouping. These are sensitivity analyses, not proof that either grouping is independent. Table 7 reports baseline-ranking point summaries; the small group count prevents an equivalence conclusion.

Variant	Grouping	Calls	Groups	Vision ρ	Language ρ
Frozen	Recorded episodes	400	16	0.929 [0.905, 0.940]	0.862 [0.835, 0.879]
Frozen	Shared-context sensitivity	400	8	0.929 [0.904, 0.947]	0.862 [0.820, 0.886]
Cascade	Recorded episodes	400	16	0.713 [0.696, 0.738]	0.910 [0.896, 0.920]
Cascade	Shared-context sensitivity	400	8	0.713 [0.693, 0.763]	0.910 [0.894, 0.926]

Table 6: Saved backbone-randomization variants. Shared-context grouping joins nominal task/episode keys across recorded seeds as a dependence sensitivity; it does not authenticate shared inputs. Intervals are conditional on the stated grouping. Learned encoders/adaptors remain, and sidecar reuse and language-baseline deviations remain unresolved.

Task	Calls	Groups	Black/gray	Black/blur	Gray/blur
PickCube	30	2	0.825	0.539	0.455
StackCube	30	2	0.814	0.546	0.496

Table 7: Saved baseline-ranking correlations. Only two nominal episode groups are available per task. Interval output is retained in the analysis artifact as a descriptive resampling diagnostic, but is not presented here as calibrated uncertainty. Positive correlation does not establish baseline interchangeability.

C Saved Integration-Budget Cohorts

Table 8 replaces the previous table of unrecovered and duplicated constants with the reconstructible later cohorts. The norm-selected population and occurrence rule are retained explicitly for this historical sensitivity. These calls are not paired with the earlier $m = 64$ main pass on identical observations, baselines, runtime and noise. Truncating trajectories to the same call depth does not create such pairs. An integration-budget effect must instead be evaluated on authenticated fixed contexts, with precision and quadrature controlled.

Task	Seed	Calls	Groups	Vision residual $\leq 3\%$ (%)
PegInsertion	42	189	8	64.0 [50.3, 76.9]
PegInsertion	142	160	7	76.2 [67.1, 85.0]
PickYCB	42	194	8	73.7 [62.1, 85.5]
PickYCB	142	195	8	71.8 [60.9, 83.1]
PegInsertion (conflicts excluded)	142	142	7	75.4 [63.8, 85.0]

Table 8: Reconstructible $m = 128$ cohorts only. These conditional intervals describe recorded cohorts, not convergence under paired contexts. The final row removes every conflicting repeated key. Missing original PickCube/StackCube subsets are not reconstructed or replaced with constants.